

CHEMAL, Russian Federation

WMO: 360580

Lat: 51.388N

Lon: 86.021E

Elev: 1376

StdP: 13.98

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.2	-15.1	-24.8	1.5	-21.2	-18.6	2.1	-14.6	22.9	18.5	19.0	17.3	0.2	340	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	88.1	66.2	84.6	65.1	81.4	63.6	68.8	83.1	67.0	80.6	65.4	77.7	4.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	94.5	73.1	62.4	88.7	71.6	60.9	84.1	70.0	33.9	83.0	32.4	80.7	31.1	77.8	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	9.8	8.7	-25.4	95.9	6.6	3.5	-30.2	98.5	-34.0	100.5	-37.7	102.5	-42.5	105.0		
(5)			-25.2	72.0	6.4	2.5	-29.9	73.8	-33.6	75.3	-37.2	76.7	-41.9	78.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	10.4	16.2	31.1	44.6	54.6	64.4	67.4	63.9	53.8	42.7	27.5	15.5		
	DBStd	21.85	12.59	12.46	11.03	9.42	8.32	6.40	5.18	6.25	7.63	8.02	12.25	12.10		
	HDD50	5065	1228	947	588	206	51	1	0	1	46	251	678	1068		
	HDD65	8963	1693	1367	1051	611	333	86	35	94	342	692	1126	1533		
	CDD50	1831	0	0	2	45	194	434	538	432	160	24	2	0		
	CDD65	254	0	0	0	0	11	68	108	61	6	0	0	0		
	CDH74	3144	0	0	0	51	325	852	1049	733	133	1	0	0		
	CDH80	998	0	0	0	9	92	268	355	243	31	0	0	0		
(14) Wind	WSAvg	3.0	4.1	3.0	2.5	2.7	2.9	2.7	2.3	2.3	2.5	3.0	3.4	4.2		
(15) Precipitation	PrecAvg	21.3	0.4	0.4	0.7	1.6	2.6	3.4	4.1	3.5	2.0	1.2	0.9	0.5		
	PrecMax	26.5	0.8	0.9	1.5	3.8	6.3	5.7	6.1	5.9	4.7	3.0	2.6	1.2		
	PrecMin	15.0	0.1	0.0	0.2	0.4	1.2	1.5	2.7	1.8	0.7	0.4	0.2	0.0		
	PrecStd	2.8	0.2	0.2	0.4	0.9	1.1	1.0	0.9	1.1	1.0	0.7	0.6	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	48.2	64.9	80.6	88.1	90.3	93.2	93.0	84.3	70.2	59.4	48.0		
		MCWB	36.8	38.0	47.4	55.6	63.4	65.9	68.9	68.5	61.6	52.7	45.9	37.7		
	2%	DB	38.5	42.6	58.1	72.9	82.0	86.5	87.6	86.3	77.6	64.9	53.8	41.6		
		MCWB	31.1	33.9	43.7	52.4	60.3	65.2	67.3	66.3	59.2	50.3	42.7	33.0		
	5%	DB	33.3	37.9	53.1	67.5	77.3	83.0	83.8	81.9	72.4	60.4	48.4	37.6		
		MCWB	27.3	30.7	41.4	50.2	57.3	64.1	66.4	64.7	57.0	48.2	39.0	30.3		
	10%	DB	28.3	33.2	47.6	62.4	71.9	79.1	80.1	77.8	67.7	56.0	42.9	32.8		
		MCWB	24.0	27.6	37.9	47.6	55.5	62.7	64.8	62.9	55.0	45.4	35.8	27.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.2	38.5	48.2	56.9	66.2	69.5	73.0	70.8	63.7	54.1	46.8	37.5		
		MCDB	46.0	47.2	62.7	75.1	84.7	84.5	87.8	86.9	80.6	66.3	58.4	47.3		
	2%	WB	31.6	34.4	44.6	53.9	61.8	67.3	69.3	67.8	60.5	51.3	42.9	33.3		
		MCDB	37.7	41.8	56.7	69.5	77.1	81.7	82.8	82.9	74.6	62.7	52.7	40.6		
	5%	WB	27.4	31.0	41.7	51.2	59.0	65.5	67.5	65.9	58.3	48.8	39.6	30.6		
		MCDB	32.7	37.2	52.1	64.6	73.6	78.9	80.4	78.9	70.0	58.9	47.2	37.1		
	10%	WB	23.9	27.8	38.4	48.7	56.7	63.9	65.8	64.1	56.0	46.3	36.3	27.7		
		MCDB	28.3	33.1	46.9	60.8	68.9	76.0	77.1	75.1	66.0	55.0	42.4	32.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	23.3	25.7	24.9	25.4	23.4	21.9	22.6	23.1	19.8	16.1	14.5		
		MCDBR	18.6	23.8	30.9	36.0	35.9	31.7	29.6	31.7	32.3	28.1	22.0	18.4		
	5% WB	MCWBR	12.8	16.6	17.8	18.1	16.8	13.2	11.7	13.3	14.8	14.9	13.0	12.5		
		MCWBR	18.4	23.5	29.2	32.3	30.8	27.0	26.3	28.3	28.6	26.4	20.9	17.8		
(40) Clear-Sky Solar Irradiance	taub		0.290	0.327	0.388	0.450	0.437	0.432	0.449	0.425	0.380	0.366	0.324	0.293		
	taud		2.073	2.009	1.942	1.964	2.098	2.210	2.181	2.224	2.286	2.191	2.110	2.041		
	Ebn at Noon		219	242	248	248	259	262	254	254	253	227	201	192		
	Edh at Noon		25	36	47	52	48	43	44	40	33	30	24	22		
(44) All-Sky Solar Radiation	RadAvg		423	763	1213	1461	1675	1852	1698	1516	1137	654	384	304		
	RadStd		40	47	96	111	161	170	102	111	96	75	36	32		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page